CVICU TRAINER

CVICU Badge Buddies

How to make them: print at **100% / Actual size** (not "fit to page"), ideally on card stock. Cut on the solid lines, **fold on the dashed line** so the front and back face out, then laminate or slip into a badge sleeve. Each card is ID-badge size (3.375 × 2.125 in).

fold

CVICU BADGE BUDDY — 12-lead: walls & arteries — front

I HIGH LATERAL LCx · diagonal (LAD)	**aVR** LEFT MAIN / 3-VESSEL STE aVR + diffuse ST↓	**V1** SEPTAL LAD (septal)	**V4** ANTERIOR LAD
II INFERIOR RCA (~80%) or LCx	**aVL** HIGH LATERAL LCx · diagonal (LAD)	**V2** SEPTAL LAD (septal)	**V5** LOW LATERAL LCx · LAD
III INFERIOR RCA	**aVF** INFERIOR RCA or LCx	**V3** ANTERIOR LAD	**V6** LOW LATERAL LCx

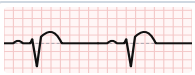
↔ **Reciprocal ST↓ supports STEMI:** inferior STE ↔ ST↓ in I/aVL · lateral STE ↔ ST↓ in II, III, aVF

\+ **Posterior:** ST↓ + tall R in V1–V3 → record V7–V9 (LCx/RCA). **RV infarct:** inferior STEMI + low BP → V4R STE (proximal RCA): fluids, avoid nitrates

CVICU BADGE BUDDY — STEMI mimics — back



LBBB / V-paced V1
Wide QRS; ST goes **opposite** the QRS. **Concordant** STE or ST↓ V1–V3 = STEMI (Sgarbossa)



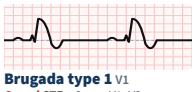
LVH V2
Very deep S V1–V3 with STE; tall R V5–V6 with **strain** (ST↓, T inversion)



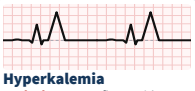
Early repolarization V3
Young; J-point **notch**, concave STE V2–V5, tall T; no reciprocal ST↓; stable



Pericarditis II
Diffuse concave STE + **PR depression** (PR↑ in aVR); pain better sitting up; rub



Brugada type 1 V1
Coved STE ≥ 2 mm V1–V2 → inverted T; RBBB-like; VF risk; fever unmasks



Hyperkalemia
Peaked narrow T, flat P, wide QRS; can raise ST in V1–V2. Check K^+

Also: LV aneurysm (old Q waves + persistent STE) · Takotsubo (looks like anterior STEMI: needs cath). **STEMI clues:** regional, reciprocal, dynamic. Compare old ECGs; when in doubt, treat as STEMI and call.

fold

CVICU BADGE BUDDY — Hemodynamics — front

NORMAL VALUES

MAP	70–105 mmHg
CVP / RAP	2–6 mmHg
PA (S/D)	15–30 / 8–15 mmHg
PAWP	6–12 mmHg
CO	4–8 L/min
CI	2.5–4 L/min/m^2
SV	60–100 mL/beat
SVR	800–1200 dyn·s/cm^5
PVR	< 250 dyn·s/cm^5
SvO_2 / $ScvO_2$	60–80% / ≥ 70%
Lactate	< 2 mmol/L

FORMULAS

MAP = (SBP + 2 × DBP) ÷ 3
CO = HR × SV
CI = CO ÷ BSA
SVR = 80 × (MAP − CVP) ÷ CO
PVR = 80 × (mPAP − PAWP) ÷ CO
CPP = DBP − PAWP

GET GOOD NUMBERS

- Level at the phlebostatic axis; zero
- Read at end-expiration
- Square-wave test: 1–2 bounces
- Trend values; treat the patient

CVICU BADGE BUDDY — Shock profiles — back

	CVP	PAWP	CI	SVR	SvO_2
Hypovolemic / bleeding	↓	↓	↓	↑	↓
Cardiogenic (LV)	↑	↑	↓	↑	↓
RV failure	↑	N / ↓	↓	↑	↓
Tamponade	↑	↑ (≈ CVP)	↓	↑	↓
Massive PE	↑	N / ↓	↓	↑	↓
Distributive (sepsis, vasoplegia)	↓ / N	↓ / N	↑ / N	↓	↑ / N

After cardiac surgery: CVP ≈ PAD ≈ PAWP + falling CI + chest tube output that suddenly stops → **tamponade**: notify the surgeon now. High CI + low SVR soon after bypass → vasoplegia.

Cold & wet (low CI, high PAWP) → inotrope + diuretic · **Warm** (high CI) with low SVR → vasopressor · **Low CVP + PAWP** → volume or blood. Follow orders and unit protocols.

fold

CVICU BADGE BUDDY — Temporary pacing · VVI — front

V Paces the ventricle	**V** Senses the ventricle	**I** Inhibits when it senses a beat

EVERY SHIFT

- Mode **VVI**; rate; **lock** on
- **Capture threshold** (if stable); output 2–3 × threshold
- Sensitivity: set in OR / cath lab
- **Underlying rhythm** (unless MD defers or unstable)
- Battery, connections, wires secured
- TVP: **cm at the hub**; balloon syringe **locked**

DOCUMENT

- Mode · rate
- Capture threshold · output
- Underlying rhythm

SENSITIVITY FENCE

- Higher mV = **less** sensitive
- Lower mV = **more** sensitive

Epicardial: ventricular wires only; cap & keep dry; out with the chest tubes. Femoral TVP: leg straight, don't bend at the waist, check pulses.

CVICU BADGE BUDDY — Pacing troubleshooting — back

LOSS OF CAPTURE

Spike, no QRS after it
Patient first · ↑ output · check battery & every connection · turn onto **left side** · notify

UNDERSENSING

Spikes on QRS or T waves (R-on-T risk)
Turn mV **down** (more sensitive) until it senses · document · notify MD

OVERSENSING

Pauses; SENSE light without real beats
Remove cause · turn mV **up** (less sensitive) · document · notify MD

NO OUTPUT

No spikes, no PACE light below set rate
Power, connections, new battery or pacer · ZOLL pads / TCP if no rhythm

Notify the provider: loss of capture · sudden ↑ in capture threshold · failure to sense despite troubleshooting · hemodynamic instability · bleeding at the insertion site. **Unstable?** Adjust the settings as needed, then notify the MD.

Educational reference. Follow your unit's protocols, provider orders and device instructions. © 2026 Alexander Bohn · CVICU Trainer